

Problem Set 2 - IO

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Question 1

Exploring the Data

```
# Load necessary libraries  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)  
library(readr)  
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
library(here)
```

here() starts at /Users/tate/SchoolWork

```
library(kableExtra)
```

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

group_rows

```
library(modelsummary)
# Load the dataset
data <- read_csv(here("Y2S1/I0/ps2/data", "prod_data_all.csv"))
```

Rows: 57748 Columns: 9

```
-- Column specification -----
Delimiter: ","
chr (2): industry, country
dbl (7): id, year, Y, Y2, L, K, M

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Clean dataset to harmonize year discrepancy - 1981 = 81
data <- data %>%
  mutate(year = ifelse(year < 100, year + 1900, year))

# Display the first few rows of the dataset
head(data)
```

```
# A tibble: 6 x 9
   id year    Y    Y2    L    K    M industry country
<dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <chr>    <chr>
1 10250 1981 296. 113. 2.86 15.5 183. metal    col
```

2	10732	1989	4733.	2566.	86.2	470.	2167.	metal	col
3	10732	1990	4800.	2608.	73.6	644.	2192.	metal	col
4	10962	1982	7090.	1836.	37.9	5538.	5254.	metal	col
5	10962	1983	4942.	1343.	40.5	6314.	3599.	metal	col
6	11192	1981	1813.	1311.	24.0	1723.	502.	metal	col

1.1

```
# Summary Statistics of the data
data %>%
  group_by(country, industry) %>%
  summarize(
    total_obs_sect = n_distinct(id),
    .groups = "drop"
  ) %>%
  ungroup() %>%
  arrange(country, industry)
```

```
# A tibble: 10 x 3
  country industry total_obs_sect
  <chr>    <chr>          <int>
1 chile   apparel           856
2 chile   food             2584
3 chile   metal            1038
4 chile   text              836
5 chile   wood              867
6 col     apparel           744
7 col     food              908
8 col     metal             632
9 col     text              475
10 col    wood             173
```

```
data %>%
  group_by(year, industry) %>%
  summarize(
    n_obs_t = n_distinct(id),
    .groups = "drop"
  ) %>%
  ungroup() %>%
  arrange(year, industry) %>%
  print()
```

```
# A tibble: 90 x 3
  year industry n_obs_t
<dbl> <chr>    <int>
1  1979 apparel     354
2  1979 food      1171
3  1979 metal      373
4  1979 text       391
5  1979 wood       340
6  1980 apparel     317
7  1980 food      1067
8  1980 metal      350
9  1980 text       356
10 1980 wood       306
# i 80 more rows
```

Chile: 1. Food - 2584, 2. Metal - 1038, 3. Wood - 867, 4. Apparel - 856, 5. Textile - 836

Colombia: 1. Food - 908, 2. Apparel - 744, 3. Metal - 632, 4. Textile - 475, 5. Wood - 173

Apparel Over Time - 1979: 354 → 1996: 292 - Peak at 1981 - 979

Food Over Time - 1979: 1171 → 1996: 1178 - Peak at 1981 - 1951

Metal Over Time - 1979: 373 → 1996: 443 - Peak at 1981 - 890

Textiles Over Time - 1979: 391 → 1996: 304 - Peak at 1981 - 752

Wood Over Time - 1979: 340 → 1996: 290 - Peak at 1981 - 465

1.2

```
# Creation of Investment Variable with 10 per depreciation rate
data <- data %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    dep      = 0.10,
    k_next   = lead(K),
    i_raw    = k_next - (1 - dep) * K,
    log_i    = ifelse(i_raw > 0, log(i_raw), NA_real_),
    log_L    = ifelse(L > 0, log(L), NA_real_),
    log_K    = ifelse(K > 0, log(K), NA_real_),
    log_Y2   = ifelse(Y2 > 0, log(Y2), NA_real_)
```

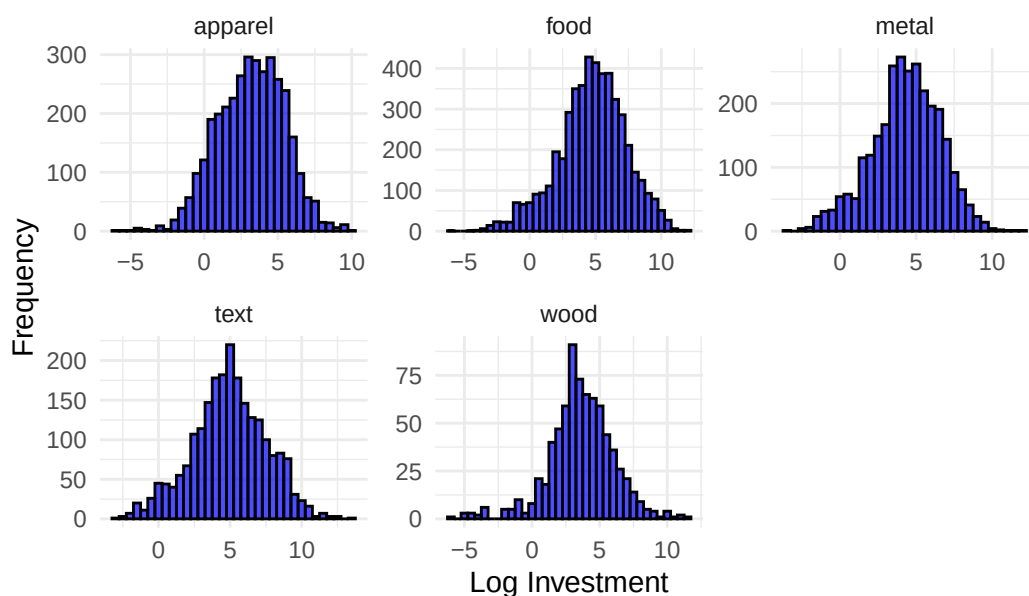
```
) %>%  
ungroup()
```

Warning: There were 2813 warnings in `mutate()`.
The first warning was:
i In argument: `log\_i = ifelse(i\_raw > 0, log(i\_raw), NA\_real\_)`.
i In group 3: `id = 54`.
Caused by warning in `log()`:
! NaNs produced
i Run `dplyr::last\_dplyr\_warnings()` to see the 2812 remaining warnings.

```
data_col <- data %>% filter(country == "col")  
data_chi <- data %>% filter(country == "chile")  
  
# Visualize with Histogram for each industry  
ggplot(data_col, aes(x = log_i)) +  
  geom_histogram(binwidth = 0.5, fill = "blue", color = "black", alpha = 0.7) +  
  facet_wrap(~ industry, scales = "free") +  
  labs(title = "Histogram of Log Investment by Industry",  
        x = "Log Investment",  
        y = "Frequency") +  
  theme_minimal()
```

Warning: Removed 3768 rows containing non-finite outside the scale range
(`stat\_bin()`).

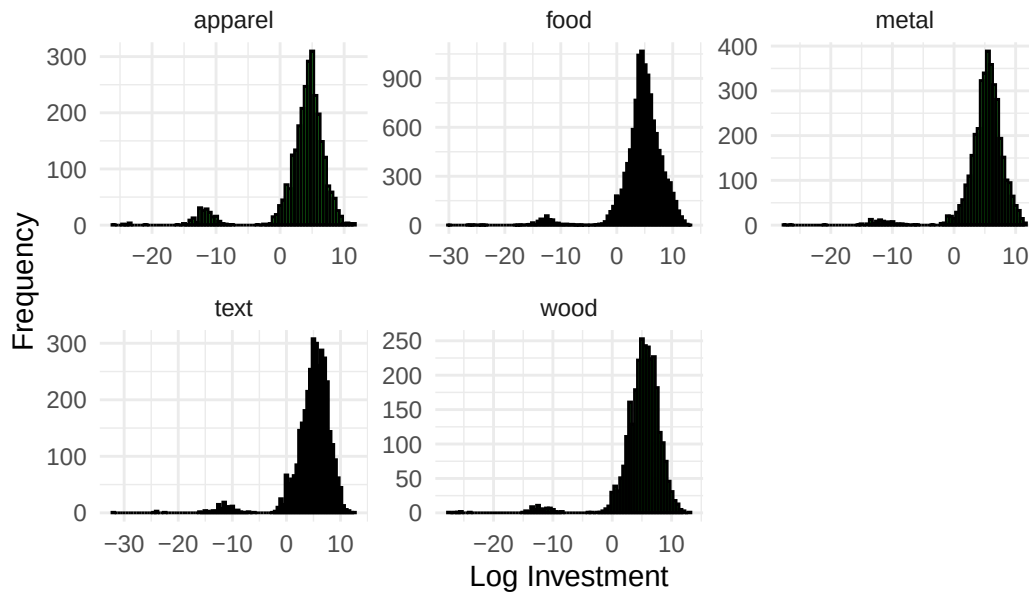
Histogram of Log Investment by Industry



```
ggplot(data_chi, aes(x = log_i)) +
  geom_histogram(binwidth = 0.5, fill = "green", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Investment by Industry",
       x = "Log Investment",
       y = "Frequency") +
  theme_minimal()
```

Warning: Removed 14540 rows containing non-finite outside the scale range
(`stat\_bin()`).

Histogram of Log Investment by Industry



From these graphs, it seems that most industries have similar levels of investment, though textiles, wood, and metal appear to have slightly higher investment levels on average, while apparel is lower.

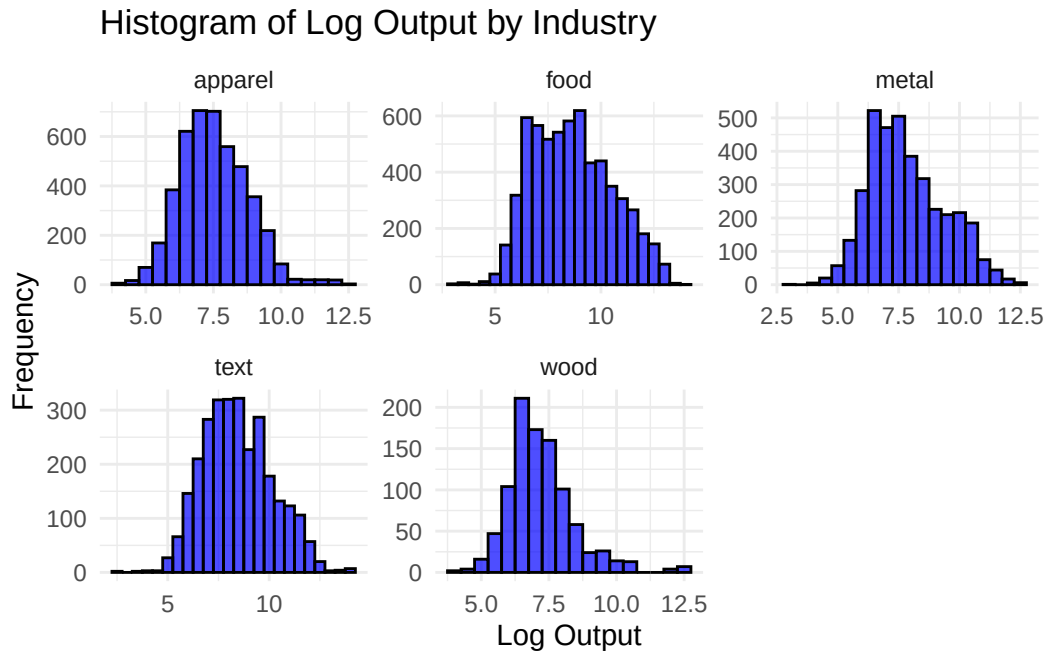
1.3

```
data_col <- data_col %>%
  group_by(industry) %>%
  mutate(
    log_Y = log(Y),
  ) %>%
  ungroup()

data_chi <- data_chi %>%
  group_by(industry) %>%
  mutate(
    log_Y = log(Y),
  ) %>%
  ungroup()

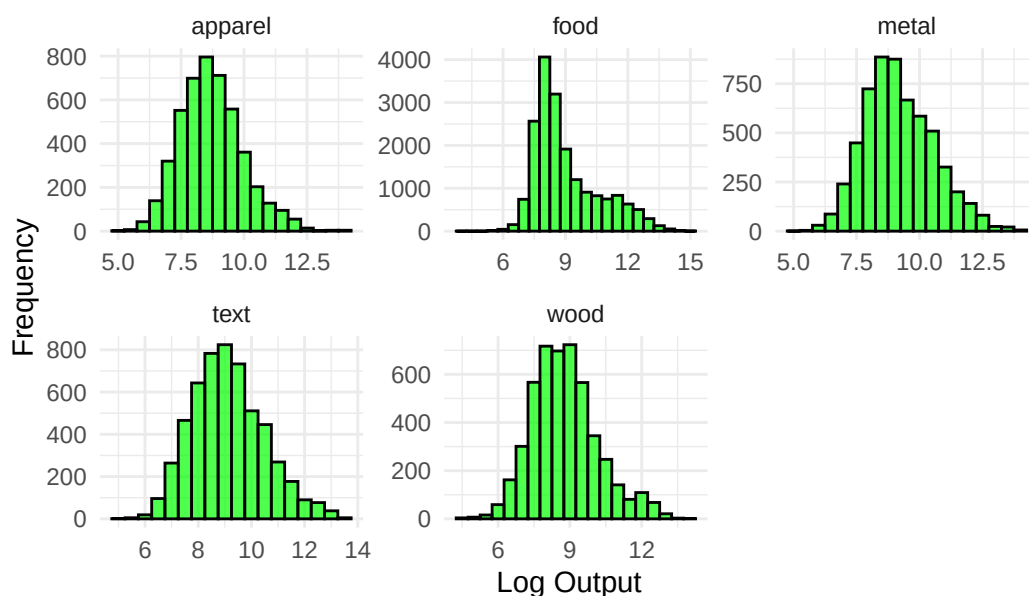
# Visualize with Histogram for each industry
ggplot(data_col, aes(x = log_Y)) +
```

```
geom_histogram(binwidth = 0.5, fill = "blue", color = "black", alpha = 0.7) +
facet_wrap(~ industry, scales = "free") +
labs(title = "Histogram of Log Output by Industry",
      x = "Log Output",
      y = "Frequency") +
theme_minimal()
```



```
ggplot(data_chi, aes(x = log_Y)) +
geom_histogram(binwidth = 0.5, fill = "green", color = "black", alpha = 0.7) +
facet_wrap(~ industry, scales = "free") +
labs(title = "Histogram of Log Output by Industry",
      x = "Log Output",
      y = "Frequency") +
theme_minimal()
```


Histogram of Log Output by Industry



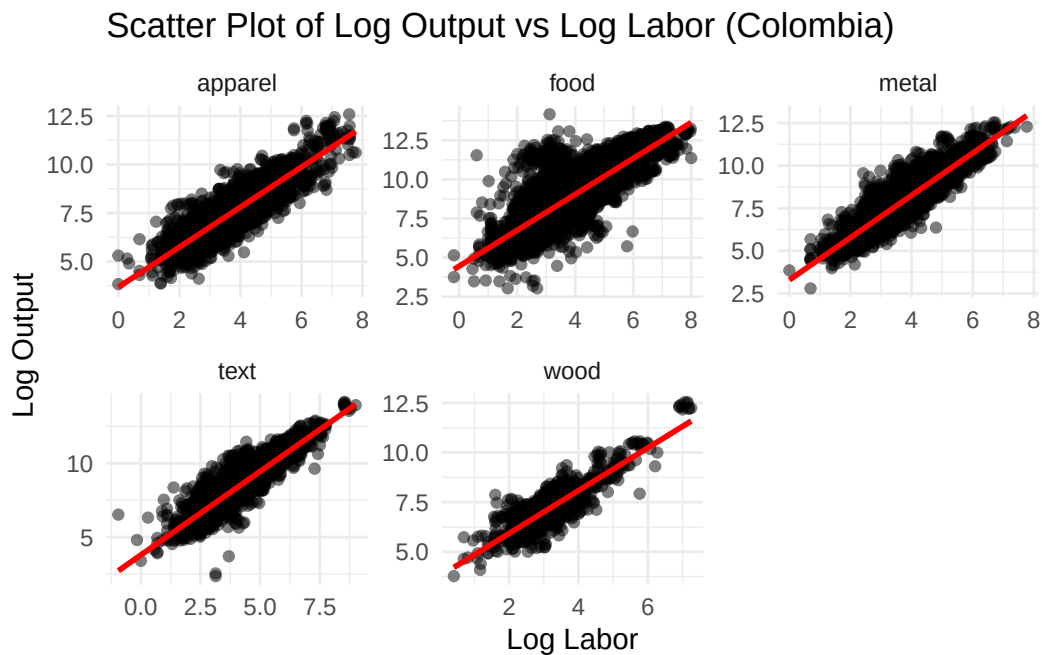
1.4

```
# Scatter plot of log_Y and log_L for each sector in each country
data_col <- data_col %>%
  group_by(industry) %>%
  mutate(
    log_L = log(L),
    log_K_ind_col = log(K),
    log_L_ind_col = log(L)
  ) %>%
  ungroup()
data_chi <- data_chi %>%
  group_by(industry) %>%
  mutate(
    log_L = log(L),
    log_K_ind_chi = log(K),
    log_L_ind_chi = log(L)
  ) %>%
  ungroup()

ggplot(data_col, aes(x = log_L, y = log_Y)) +
  geom_point(alpha = 0.5) +
```

```
geom_smooth(method = "lm", color = "red") +
facet_wrap(~ industry, scales = "free") +
labs(title = "Scatter Plot of Log Output vs Log Labor (Colombia)",
      x = "Log Labor",
      y = "Log Output") +
theme_minimal()
```

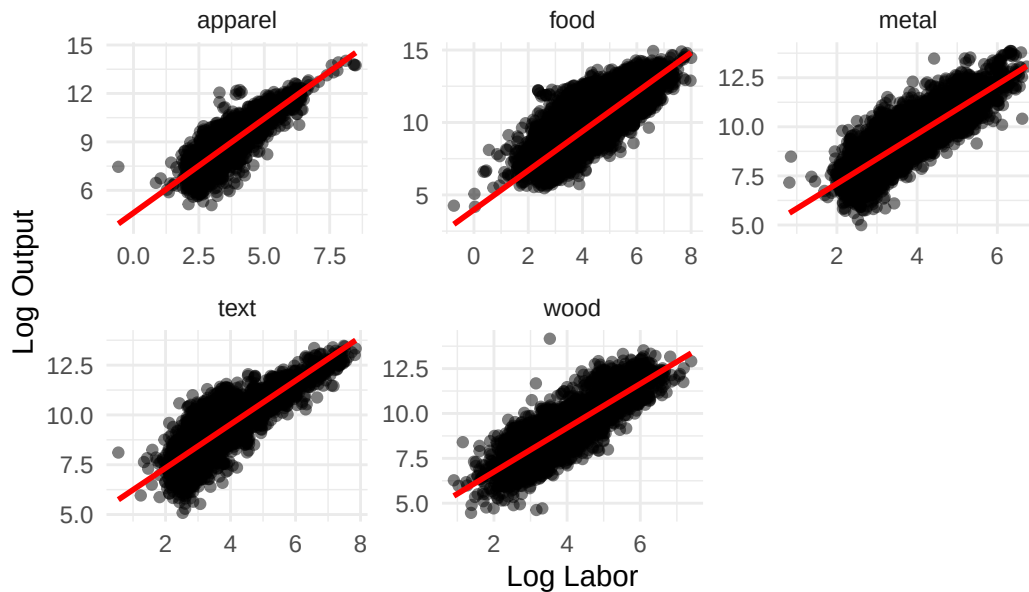
`geom\_smooth()` using formula = 'y ~ x'



```
ggplot(data_chi, aes(x = log_L, y = log_Y)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "red") +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Scatter Plot of Log Output vs Log Labor (Chile)",
        x = "Log Labor",
        y = "Log Output") +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

Scatter Plot of Log Output vs Log Labor (Chile)



Question 2

2.1

The estimating equation is as follows:

$$\log(Y_{it}) = \alpha_0 + \omega_{it} + \epsilon_{it} + \alpha_L \log(L_{it}) + \alpha_K \log(K_{it})$$

To get here, we take the log of:

$$Y_{it} = e^{\alpha_0 + \omega_{it} + \epsilon_{it}} L_{it}^{\alpha_L} K_{it}^{\alpha_K}$$

Where Y_{it} is the value added output of firm i at time t , L_{it} is labor input and K_{it} is capital input. The ω_{it} term is heterogeneous productivity shocks, varying across firms and years. As discussed in class, we can use OLS to estimate, but we will have some issues with endogeneity via ω_{it} showing up in the labor FOC.

```
# Estimate homogenous parameteres using OLS for each country
ols_col <- lm(log(Y2) ~ log_L + log_K, data = data_col)

ols_chi <- lm(log(Y2) ~ log_L + log_K, data = data_chi)
```

	Colombia	Chile
(Intercept)	2.186*** (0.018)	2.351*** (0.020)
log_L	0.831*** (0.006)	0.887*** (0.007)
log_K	0.273*** (0.004)	0.303*** (0.004)
Num.Obs.	18 083	39 665
R2	0.823	0.660
R2 Adj.	0.823	0.660
AIC	291 626.5	725 786.1
BIC	291 657.7	725 820.5
Log.Lik.	−18 545.704	−54 890.486
F	42 113.270	38 426.063
RMSE	0.67	0.97
* p < 0.1, ** p < 0.05, *** p < 0.01		

```

modelsummary(
  list(Colombia = ols_col, Chile = ols_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

```

```

# Estimate varying parameters across industries using OLS for each country

ols_ind_col <- lm(log(Y2) ~ industry + log_L:industry + log_K:industry,
  data = data_col)
ols_ind_chi <- lm(log(Y2) ~ industry + log_L:industry + log_K:industry,
  data = data_chi)

modelsummary(
  list(Colombia_Ind = ols_ind_col, Chile_Ind = ols_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

```

As can be seen in the regression outputs, when we add heterogeneity across industries, the estimates for labor and capital change slightly. However, we can now ascertain differences in productivity across industries via the industry coefficients.

	Colombia_Ind	Chile_Ind
(Intercept)	2.510*** (0.040)	2.865*** (0.063)
industryfood	−0.272*** (0.049)	−0.915*** (0.068)
industrymetal	−0.536*** (0.058)	0.131 (0.082)
industrytext	−0.294*** (0.059)	0.156* (0.083)
industrywood	0.049 (0.089)	−0.377*** (0.085)
industryapparel × log_L	0.921*** (0.013)	1.055*** (0.021)
industryfood × log_L	0.754*** (0.010)	0.896*** (0.010)
industrymetal × log_L	0.998*** (0.016)	0.994*** (0.018)
industrytext × log_L	0.858*** (0.015)	0.874*** (0.017)
industrywood × log_L	0.974*** (0.030)	1.016*** (0.020)
industryapparel × log_K	0.133*** (0.009)	0.150*** (0.013)
industryfood × log_K	0.321*** (0.007)	0.334*** (0.005)
industrymetal × log_K	0.208*** (0.010)	0.232*** (0.009)
industrytext × log_K	0.257*** (0.010)	0.243*** (0.010)
industrywood × log_K	0.117*** (0.018)	0.174*** (0.010)
Num.Obs.	18 083	39 665
R2	0.831	0.689
R2 Adj.	0.831	0.689
AIC	290 826.2	722 221.0
BIC	290 951.0	722 358.4
Log.Lik.	−18 133.547	−53 095.913
F	6352.788	6275.729
RMSE	0.66	0.92

* p < 0.1, ** p < 0.05, *** p < 0.01

	Colombia	Chile
(Intercept)	2.282*** (0.023)	2.521*** (0.026)
log_L	0.828*** (0.007)	0.844*** (0.009)
log(K)	0.242*** (0.006)	0.299*** (0.005)
log_i	0.031*** (0.004)	0.017*** (0.002)
Num.Obs.	14 315	25 125
R2	0.836	0.689
R2 Adj.	0.836	0.689
AIC	232 325.5	471 395.1
BIC	232 363.3	471 435.7
Log.Lik.	-13 820.998	-33 699.484
F	24 327.769	18 593.859
RMSE	0.64	0.93

* p < 0.1, ** p < 0.05, *** p < 0.01

2.2

```
form_hom <- as.formula("log(Y2) ~ log_L + log(K) + log_i")

ols_inv_col <- lm(form_hom, data = data_col)
ols_inv_chi <- lm(form_hom, data = data_chi)

modelsummary(
  list(Colombia = ols_inv_col, Chile = ols_inv_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

```
form_ind <- as.formula("log(Y2) ~ industry + log_L:industry + log_K:industry + log_i")
ols_inv_ind_col <- lm(form_ind, data = data_col)
ols_inv_ind_chi <- lm(form_ind, data = data_chi)

modelsummary(
  list(Colombia_Ind = ols_inv_ind_col, Chile_Ind = ols_inv_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
```

```

    output = "kableExtra"
  )

```

When we add investment as a regressor, we see that the coefficients on labor and capital change slightly again. The investment coefficient is positive and significant across all specifications, indicating that higher investment is associated with higher value added output. Industry level heterogeneity remains mostly significant as well.

Comparing to the results from 2.1, inclusion of investment seems to improve model fit slightly, as seen in the adjusted R-squared values, suggesting that investment is a factor worth considering when modeling firm output.

2.3

```

# First stage: estimate log_y2 = log_l + phi(log_k) s.t. phi = alpha_0 +
# alpha_k*log_k + a(i*t*k)^2 + b(i*t*k) + c

data_col <- data_col %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    log_k_next = lead(log_K),
    log_y2_next = lead(log_Y2)
  ) %>%
  ungroup()
data_chi <- data_chi %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    log_k_next = lead(log_K),
    log_y2_next = lead(log_Y2)
  ) %>%
  ungroup()

# Define a function to perform the first stage regression and get phi_hat
first_stage <- function(data) {
  data <- data %>%
    mutate(
      k_sq = log_K^2,
      it_k = as.numeric(factor(paste(id, year))) * log_K,
      it_k_sq = it_k^2,

```

	Colombia_Ind	Chile_Ind
(Intercept)	2.583*** (0.045)	3.122*** (0.080)
industryfood	−0.241*** (0.053)	−1.016*** (0.085)
industrymetal	−0.510*** (0.064)	0.071 (0.103)
industrytext	−0.355*** (0.065)	0.200* (0.102)
industrywood	0.027 (0.099)	−0.504*** (0.107)
log_i	0.027*** (0.004)	0.019*** (0.002)
industryapparel × log_L	0.923*** (0.015)	1.004*** (0.026)
industryfood × log_L	0.769*** (0.011)	0.842*** (0.012)
industrymetal × log_L	0.983*** (0.017)	0.945*** (0.022)
industrytext × log_L	0.839*** (0.017)	0.848*** (0.020)
industrywood × log_L	0.958*** (0.034)	0.978*** (0.025)
industryapparel × log_K	0.107*** (0.011)	0.144*** (0.017)
industryfood × log_K	0.282*** (0.008)	0.336*** (0.006)
industrymetal × log_K	0.185*** (0.012)	0.221*** (0.012)
industrytext × log_K	0.246*** (0.011)	0.211*** (0.012)
industrywood × log_K	0.105*** (0.020)	0.178*** (0.013)
Num.Obs.	14 315	25 125
R2	0.843	0.715
R2 Adj.	0.843	0.715
AIC	231 717.4	469 239.8
BIC	231 846.1	469 378.1
Log.Lik.	−13 504.957	−32 609.873
F	5123.978	4205.456
RMSE	0.62	0.89

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$


```

    inv_sq = log_i^2,
    k_inv = log_K * log_i,
    k_inv_sq = k_inv^2,
    rhs = cbind(log_L, log_K, k_sq, log_i, inv_sq, k_inv, k_inv_sq)
  )
  fs_model <- lm(log_Y2 ~ rhs, data = data)
  data$phi_hat <- predict(fs_model, newdata = data)
  return(data)
}
data_col <- first_stage(data_col)
data_chi <- first_stage(data_chi)

# Second stage: estimate  $y_{2,t+1} - \alpha_l l_{t+1} = \alpha_0 + \alpha_k k_{t+1}$ 
# +  $g(\phi_{hat,t} - \alpha_k k_t - \alpha_0) + \text{error}$ 

form_op <- as.formula("log_y2_next - log_L ~ log_k_next + I(phi_hat - log_K)")
op_col <- lm(form_op, data = data_col)
op_chi <- lm(form_op, data = data_chi)

modelsummary(
  list(Colombia = op_col, Chile = op_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

```

```

# Estimate varying parameters across industries using OP for each country
# First stage already done above
data_col <- data_col %>%
  mutate(
    industry_factor = as.factor(industry)
  )
data_chi <- data_chi %>%
  mutate(
    industry_factor = as.factor(industry)
  )
form_op_ind <- as.formula("log_y2_next - log_L:industry_factor ~ industry_factor + log_k_next")
op_ind_col <- lm(form_op_ind, data = data_col)

```

Warning in log_L:industry_factor: numerical expression has 18083 elements: only the first used

	Colombia	Chile
(Intercept)	2.337*** (0.031)	2.523*** (0.039)
log_k_next	0.162*** (0.004)	0.237*** (0.005)
I(phi_hat - log_K)	-0.142*** (0.008)	-0.114*** (0.009)
Num.Obs.	14 273	25 099
R2	0.263	0.238
R2 Adj.	0.262	0.238
AIC	30 001.0	69 617.4
BIC	30 031.3	69 649.9
Log.Lik.	-14 996.523	-34 804.709
RMSE	0.69	0.97

* p < 0.1, ** p < 0.05, *** p < 0.01

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_K:industry_factor: numerical expression has 18083 elements: only the first used

Warning in phi_hat - log_K:industry_factor: longer object length is not a multiple of shorter object length

```
op_ind_chi <- lm(form_op_ind, data = data_chi)
```

Warning in log_L:industry_factor: numerical expression has 39665 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_K:industry_factor: numerical expression has 39665 elements: only the first used

Warning in phi_hat - log_K:industry_factor: longer object length is not a multiple of shorter object length

```

modelsummary(
  list(Colombia_Ind = op_ind_col, Chile_Ind = op_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

```

Warning in log_L:industry_factor: numerical expression has 18083 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_L:industry_factor: numerical expression has 18083 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_L:industry_factor: numerical expression has 39665 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_L:industry_factor: numerical expression has 39665 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

2.4

	Colombia_Ind	Chile_Ind
(Intercept)	−0.722*** (0.112)	0.591*** (0.105)
industry_factorfood	−1.051*** (0.140)	−0.565*** (0.111)
industry_factormetal	−1.100*** (0.168)	0.430*** (0.136)
industry_factortext	−0.710*** (0.171)	−0.056 (0.137)
industry_factorwood	−0.011 (0.254)	−0.267* (0.141)
I(phi_hat - log_K × industry_factor)	0.082*** (0.007)	0.071*** (0.004)
industry_factorapparel × log_k_next	0.546*** (0.019)	0.602*** (0.015)
industry_factorfood × log_k_next	0.664*** (0.013)	0.644*** (0.006)
industry_factormetal × log_k_next	0.676*** (0.019)	0.581*** (0.011)
industry_factortext × log_k_next	0.646*** (0.018)	0.587*** (0.011)
industry_factorwood × log_k_next	0.462*** (0.037)	0.562*** (0.012)
Num.Obs.	14 273	25 099
R2	0.387	0.559
R2 Adj.	0.386	0.559
AIC	55 278.5	79 099.7
BIC	55 369.3	79 197.3
Log.Lik.	−27 627.270	−39 537.865
RMSE	1.68	1.17

* p < 0.1, ** p < 0.05, *** p < 0.01

```

first_stage_lp <- function(data) {
  data <- data %>%
    mutate(
      k_sq = log_K^2,
      it_k = as.numeric(factor(paste(id, year))) * log_K,
      it_k_sq = it_k^2,
      inv_sq = log_i^2,
      k_inv = log_K * log_i,
      k_inv_sq = k_inv^2,
      log_M = log(M),
      log_M_next = lead(log_M),
      m_sq = log_M^2,
      k_m = log_K * log_M,
      k_m_sq = k_m^2,
      m_inv = log_M * log_i,
      m_inv_sq = m_inv^2,
      rhs = cbind(log_L, log_K, k_sq, log_M, m_sq, log_i, inv_sq,
                  k_inv, k_inv_sq, k_m, k_m_sq, m_inv, m_inv_sq)
    )
  fs_model <- lm(log_Y2 ~ rhs, data = data)
  data$phi_hat <- predict(fs_model, newdata = data)
  return(data)
}

data_col_lp <- first_stage_lp(data_col)
data_chi_lp <- first_stage_lp(data_chi)

# Second stage for LP
form_lp <- as.formula("log_y2_next - log_L ~ log_k_next + log_M_next + I(phi_hat - log_K - log_i)")
lp_col <- lm(form_lp, data = data_col_lp)
lp_chi <- lm(form_lp, data = data_chi_lp)

modelsummary(
  list(Colombia = lp_col, Chile = lp_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

# Estimate varying parameters across industries using LP for each country
data_col_lp <- data_col_lp %>%
  mutate(
    industry_factor = as.factor(industry),

```

	Colombia	Chile
(Intercept)	1.537*** (0.024)	1.021*** (0.034)
log_k_next	-0.230*** (0.008)	-0.109*** (0.011)
log_M_next	0.150*** (0.006)	0.221*** (0.007)
I(phi_hat - log_K - log_M)	-0.306*** (0.009)	-0.258*** (0.012)
Num.Obs.	14 273	25 099
R2	0.445	0.310
R2 Adj.	0.445	0.310
AIC	25 950.5	67 120.7
BIC	25 988.3	67 161.3
Log.Lik.	-12 970.238	-33 555.347
RMSE	0.60	0.92

* p < 0.1, ** p < 0.05, *** p < 0.01

```

    log_M_next = lead(log_M)
  )
data_chi_lp <- data_chi_lp %>%
  mutate(
    industry_factor = as.factor(industry),
    log_M_next = lead(log_M)
  )
form_lp_ind <- as.formula("log_y2_next - log_L:industry_factor ~ industry_factor + log_k_next")
lp_ind_col <- lm(form_lp_ind, data = data_col_lp)

```

Warning in log_L:industry_factor: numerical expression has 18083 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_K:industry_factor: numerical expression has 18083 elements: only the first used

Warning in phi_hat - log_K:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_M:industry_factor: numerical expression has 18083 elements: only the first used

Warning in phi_hat - log_K:industry_factor - log_M:industry_factor: longer object length is not a multiple of shorter object length

```
lp_ind_chi <- lm(form_lp_ind, data = data_chi_lp)
```

Warning in log_L:industry_factor: numerical expression has 39665 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_K:industry_factor: numerical expression has 39665 elements: only the first used

Warning in phi_hat - log_K:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_M:industry_factor: numerical expression has 39665 elements: only the first used

Warning in phi_hat - log_K:industry_factor - log_M:industry_factor: longer object length is not a multiple of shorter object length

```
modelsummary(  
  list(Colombia_Ind = lp_ind_col, Chile_Ind = lp_ind_chi),  
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),  
  output = "kableExtra"  
)
```

Warning in log_L:industry_factor: numerical expression has 18083 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_L:industry_factor: numerical expression has 18083 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_L:industry_factor: numerical expression has 39665 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_L:industry_factor: numerical expression has 39665 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

2.5

```
# ACF Estimation for each country
acf_fs <- function(data) {
  data <- data %>%
    mutate(
      lag_log_L = lag(log_L),
      lag_log_K = lag(log_K),
      t2_lag_log_L = lag(lag_log_L),
      t2_lag_log_K = lag(lag_log_K),
      k_sq = log_K^2,
      it_k = as.numeric(factor(paste(id, year))) * log_K,
      it_k_sq = it_k^2,
      l_sq = log_L^2,
      it_l = as.numeric(factor(paste(id, year))) * log_L,
      it_l_sq = it_l^2,
      l_k = log_L * log_K,
      l_k_sq = l_k^2,
      rhs = cbind(lag_log_L, lag_log_K, t2_lag_log_L, t2_lag_log_K,
                  k_sq, l_sq, it_k, it_k_sq, it_l, it_l_sq, l_k, l_k_sq)
```


	Colombia_Ind	Chile_Ind
(Intercept)	−1.001*** (0.128)	−1.699*** (0.123)
industry_factorfood	−0.773*** (0.152)	−0.732*** (0.131)
industry_factormetal	−0.759*** (0.176)	0.439*** (0.157)
industry_factortext	−0.804*** (0.188)	−0.048 (0.161)
industry_factorwood	0.131 (0.292)	−0.538*** (0.165)
I(phi_hat - log_K × industry_factor - log_M × industry_factor)	0.173*** (0.003)	0.049*** (0.002)
industry_factorapparel × log_k_next	0.185*** (0.023)	0.243*** (0.018)
industry_factorfood × log_k_next	0.232*** (0.017)	0.222*** (0.008)
industry_factormetal × log_k_next	0.131*** (0.027)	0.214*** (0.014)
industry_factortext × log_k_next	0.136*** (0.027)	0.258*** (0.014)
industry_factorwood × log_k_next	0.171*** (0.043)	0.183*** (0.015)
industry_factorapparel × log_M_next	0.483*** (0.025)	0.604*** (0.020)
industry_factorfood × log_M_next	0.471*** (0.017)	0.662*** (0.010)
industry_factormetal × log_M_next	0.624*** (0.029)	0.624*** (0.017)
industry_factortext × log_M_next	0.628*** (0.031)	0.591*** (0.018)
industry_factorwood × log_M_next	0.445*** (0.054)	0.674*** (0.019)
Num.Obs.	14 273	25 099
R2	0.551	0.684
R2 Adj.	0.550	0.684
AIC	50 836.2	70 730.0
BIC	50 964.9	70 868.2
Log.Lik.	−25 401.123	−35 347.976
RMSE	1.43	0.99

* p < 0.1, ** p < 0.05, *** p < 0.01

	Colombia	Chile
(Intercept)	0.284*** (0.059)	0.014 (0.082)
log_L	0.116*** (0.022)	0.018 (0.031)
log_K	0.034*** (0.008)	−0.005 (0.011)
I(phi_hat)	0.867*** (0.026)	0.994*** (0.034)
Num.Obs.	18 081	39 663
R2	0.834	0.667
R2 Adj.	0.834	0.667
AIC	35 992.8	108 940.4
BIC	36 031.8	108 983.3
Log.Lik.	−17 991.388	−54 465.200
F	30 212.342	26 453.247
RMSE	0.65	0.96

* p < 0.1, ** p < 0.05, *** p < 0.01

```

)
fs_model <- lm(log_Y2 ~ rhs, data = data)
data$phi_hat <- predict(fs_model, newdata = data)
return(data)
}

data_col_acf <- acf_fs(data_col)
data_chi_acf <- acf_fs(data_chi)
# Second stage for ACF
form_acf <- as.formula("log_Y2 ~ log_L + log_K + I(phi_hat)")
acf_col <- lm(form_acf, data = data_col_acf)
acf_chi <- lm(form_acf, data = data_chi_acf)

modelsummary(
  list(Colombia = acf_col, Chile = acf_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

```

```

# Estimate varying parameters across industries using ACF for each country
data_col_acf <- data_col_acf %>%
  mutate(
    industry_factor = as.factor(industry)
  )
data_chi_acf <- data_chi_acf %>%
  mutate(
    industry_factor = as.factor(industry)
  )
form_acf_ind <- as.formula("log_Y2 ~ industry_factor + log_L:industry_factor + log_K:industry_factor")
acf_ind_col <- lm(form_acf_ind, data = data_col_acf)
acf_ind_chi <- lm(form_acf_ind, data = data_chi_acf)

modelssummary(
  list(Colombia_Ind = acf_ind_col, Chile_Ind = acf_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

```

ACF estimates

Question 3

3.1

```

# Using OP estimates from 2.3 to estimate omega = exp(phi_it - alpha_0 - alpha_k*log(K_it))

alpha_0_col <- coef(op_col)["(Intercept)"]
alpha_k_col <- coef(op_col)["log_k_next"]
data_col <- data_col %>%
  mutate(
    omega_it = exp(phi_hat - alpha_0_col - alpha_k_col * log_K)
  )
alpha_0_chi <- coef(op_chi)["(Intercept)"]
alpha_k_chi <- coef(op_chi)["log_k_next"]
data_chi <- data_chi %>%
  mutate(
    omega_it = exp(phi_hat - alpha_0_chi - alpha_k_chi * log_K)
  )
data_col <- data_col %>%

```

	Colombia_Ind	Chile_Ind
(Intercept)	1.483*** (0.154)	1.576*** (0.300)
industry_factorfood	-1.178*** (0.180)	-2.432*** (0.320)
industry_factormetal	-1.277*** (0.218)	0.786** (0.362)
industry_factortext	-1.005*** (0.209)	0.539 (0.363)
industry_factorwood	0.475 (0.301)	-1.613*** (0.398)
industry_factorapparel $\times$ log_L	0.542*** (0.057)	0.565*** (0.114)
industry_factorfood $\times$ log_L	0.032 (0.035)	-0.129*** (0.041)
industry_factormetal $\times$ log_L	0.308*** (0.060)	0.742*** (0.080)
industry_factortext $\times$ log_L	0.209*** (0.052)	0.539*** (0.075)
industry_factorwood $\times$ log_L	0.758*** (0.093)	0.079 (0.097)
industry_factorapparel $\times$ log_K	0.050*** (0.015)	-0.007 (0.038)
industry_factorfood $\times$ log_K	0.059*** (0.014)	-0.020 (0.014)
industry_factormetal $\times$ log_K	-0.011 (0.021)	0.138*** (0.031)
industry_factortext $\times$ log_K	0.023 (0.020)	0.117*** (0.029)
industry_factorwood $\times$ log_K	0.053* (0.031)	-0.160*** (0.035)
industry_factorapparel $\times$ I(phi_hat)	0.429*** (0.062)	0.544*** (0.124)
industry_factorfood $\times$ I(phi_hat)	0.893*** (0.041)	1.172*** (0.045)
industry_factormetal $\times$ I(phi_hat)	0.821*** (0.069)	0.287*** (0.089)
industry_factortext $\times$ I(phi_hat)	0.803*** (0.062)	0.391*** (0.085)
industry_factorwood $\times$ I(phi_hat)	0.259** (0.106)	1.078*** (0.109)
Num.Obs.	18 081	39 663
R2	0.839	0.695
R2 Adj.	0.838	0.695
AIC	35 489.9	105 406.9
BIC	35 653.7	105 587.2
Log.Lik.	-17 723.932	-52 682.448
F	4937.799	4764.107
RMSE	0.64	0.91

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

```

mutate(
  mean_omega_col = mean(omega_it, na.rm = TRUE),
  sd_omega_col = sd(omega_it, na.rm = TRUE)
)
data_chi <- data_chi %>%
  mutate(
    mean_omega_chi = mean(omega_it, na.rm = TRUE),
    sd_omega_chi = sd(omega_it, na.rm = TRUE)
  )

print(paste("Colombia - Mean Omega:", data_col$mean_omega_col[1], "SD Omega:", data_col$sd_omega_col[1]))

```

```
[1] "Colombia - Mean Omega: 110.32773611788 SD Omega: 310.548166738674"
```

```

print(paste("Chile - Mean Omega:", data_chi$mean_omega_chi[1], "SD Omega:", data_chi$sd_omega_chi[1]))

```

```
[1] "Chile - Mean Omega: 81.7321484462486 SD Omega: 153.754310346994"
```

As can be seen, Colombia is more productive on average, with a mean ω of ≈ 110.33 compared to Chile's mean of ≈ 81.73 . However, Colombia also has a higher standard deviation of ≈ 310.55 versus Chile's ≈ 153.75 , indicating greater variability in productivity among Colombian firms.

3.2